Cyber Defense (420-D03-AB)

Final Project & Course Procedure Manual

Cybersecurity AEC (Attestation d'études collégiales) — LEA.D8 · Competency EZ03 — To apply access control measures

Presented at the first held class — Tuesday, September 1, 2026 · Cyber.SoHo · Version 1.0

Session note: Class 1 (Saturday, Aug 29) was cancelled — its content is delivered today · every due date and weight unchanged

Tonight's Roadmap

- Why today matters: both end-of-course deliverables, explained in full
- The legal framework you accept — and sign — every session
- Section 1 — the Final Project (15%): one fully defended company
- Meet the fictional company, its people, and its systems
- The seven chapters — six of them are your labs
- The resilience assessment: the only new element
- The submission package and the complete grading checklist
- Section 2 — the Course Procedure Manual (10%)
- Templates, the documentation standard, and the AI disclosure rule
- The timeline to Class 21 — and why the workload stays manageable

25%

two deliverables — due Class 21

Session Note — and Where These 25% Sit

- Class 1 (Saturday, August 29) was cancelled — its content is delivered today
- Every deliverable, due date, and grade weight stays unchanged
- Today also: Quiz Q_01 (2.5%) and finishing LAB_01
- Both deliverables below are due Class 21 — Tuesday, October 20, 2026
- Together they are worth 25% of your course grade

The evaluation plan — 100% of the course

PROJECT (15%) + MANUAL (10%) = 25% — due Class 21

The Legal Framework (1/2) — Education Only, Accepted by All

- One purpose: teaching defensive cybersecurity — together, "the Material"
- The Material shows how to PROTECT systems — never how to attack others
- Nothing in it authorizes access to any real system, network, account, or data
- Using any part of the Material = accepting the entire Legal Notice (v1.0)
- At the start of EVERY session: sign the "Acknowledgment and Agreement" page
- Your signature creates your personal electronic record for that session
- The e-signature page and its log arrive separately on the course platform
- All 16 sections are reproduced word for word in the FULL document, Annex A

SIGN

every session — e-signature page provided separately

The Legal Framework (2/2) — The One Unbreakable Rule

- Practice ONLY inside your personal, isolated laboratory environment …
- … or on systems with explicit WRITTEN permission, obtained BEFORE you start
- Unauthorized computer access or use is a criminal offence — Criminal Code:
    - s. 342.1 — unauthorized use of a computer
    - s. 342.2 — possession of a device to obtain unauthorized use
    - s. 430(1.1) — mischief in relation to computer data
- Penalties can include imprisonment
- Outside Canada: the laws of your own country, province, or state apply to you
- Your college's policies and codes of conduct apply in addition to the law
- The instructor never asks for — and never authorizes — any illegal act

! **If an activity seems to touch a system you do not own: STOP and ask first**

STOP

& ask — before touching anything not yours

Acceptable Use — You MAY

- ✓ Practice inside your isolated lab, as defined in LAB_01
- ✓ Local virtualization (for example VirtualBox) …
- ✓ … or your own personal free-tier cloud account (AWS or Microsoft Azure)
- ✓ Other systems ONLY with explicit written permission, obtained before you start
- ✓ Use snapshots and isolated virtual networks — mistakes stay inside the lab

Cloud responsibilities

- Respect the provider's acceptable-use policies (AWS / Azure)
- Free tiers have limits — monitor them
- All cloud costs, overages, and account consequences are yours

LAB

isolated · snapshots · virtual networks

Acceptable Use — You Must NEVER

- ✗ Scan, probe, attack, intercept, monitor, or access ANY system that is not yours
- ✗ … including the college, an employer, a business, a public website, or a neighbor
- ✗ Create, install, or release malicious software outside your isolated laboratory
- ✗ Attempt password, credential, or interception attacks on any real account …
- ✗ … or against any real communication
- ✗ Use the Material, or any tool in it, to break any law or policy that applies to you
- ▪ Breaking these rules = criminal exposure + college consequences — at the same time
- **! Unsure? Stop. Ask the instructor. Then act.**

NEVER

not yours = not allowed

SECTION 1

The Final Project

Due Class 21 — Tuesday, October 20, 2026 · at the start of class

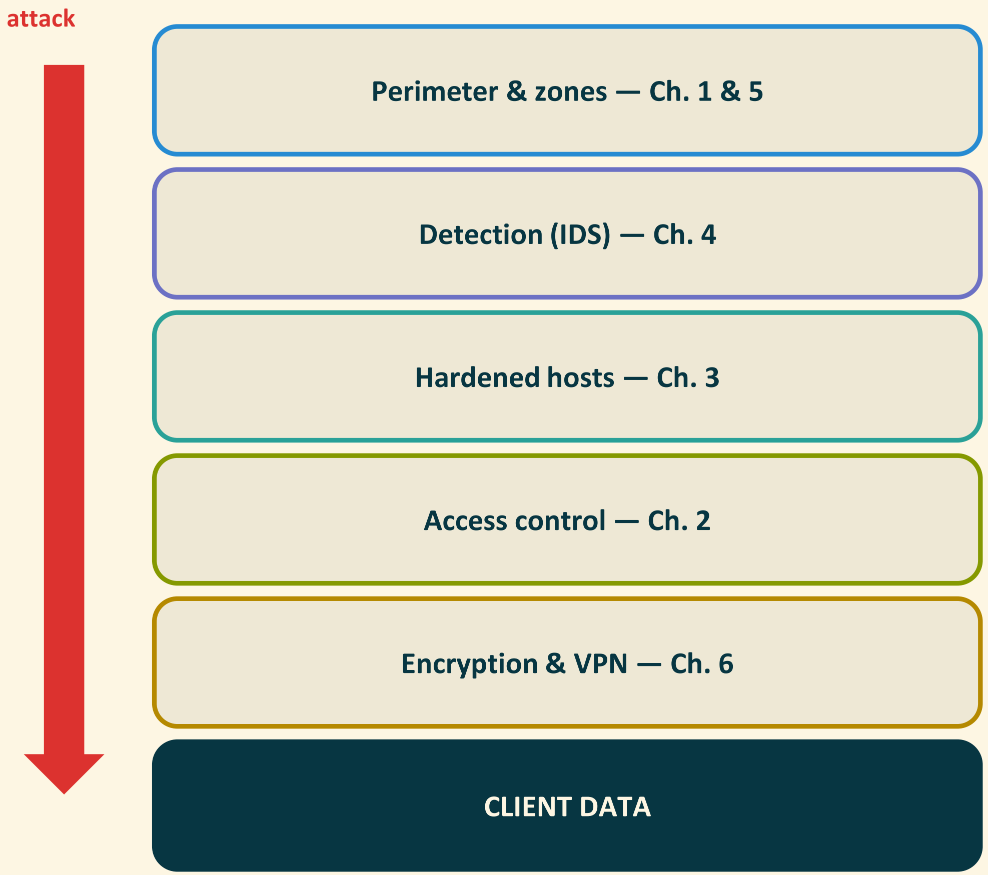
15%

of your course grade

The Goal — One Small, Fully Defended Company

- Deliver ONE small, fully defended company IT infrastructure
- Nothing new: the direct assembly of your six labs + one resilience assessment
- Keep up with the labs → the project is essentially done by Class 17
- Classes 18–21 are reserved for assembly and clean documentation
- Competency shown: EZ03 — apply access control measures, layer by layer

Defense-in-depth — every layer must be crossed



Meet the Client — NorthMaple Accounting (Fictional)

- A small accounting firm — about ten people, one office
- It holds SENSITIVE client financial records — the stakes of every choice
- One internet connection + a public company website
- An internal file server + employee workstations
- Wireless access for staff — and for visitors
- 1–2 remote workers must reach internal data securely
- You may rename it and adjust details for creativity …
- … if the role table and the minimum system list stay equivalent

≈ 10 people
one office

SENSITIVE
client financial records

Public website
+ one internet link

1–2 remote workers
via VPN only

The Role Table — Who Reaches What

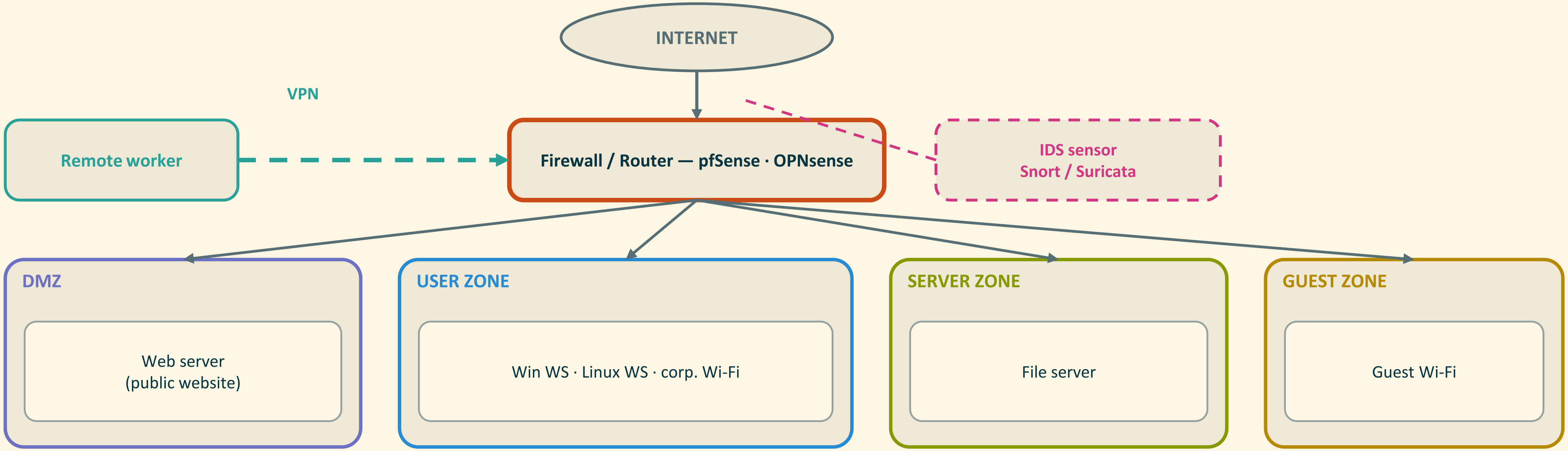
- Five roles — each reaches ONLY what its work requires (need-to-know)
- This exact table drives LAB_02 and returns in every later chapter
- Remote workers keep their role's access — but only through the VPN

Role	Count	Client records	Tech docs	Front-desk	Admin rights
Administrator (IT)	1	✓	✓	✓	Full — controlled
Accountant	3	✓	✗	✗	✗ none
Technician	2	✗	✓	✗	Workstations only
Receptionist	1	✗	✗	✓	✗ none
Remote worker	1–2	Same access as their role — reachable ONLY through the VPN			

✓ = allowed · ✗ = refused · Administrator: full, controlled access via admin group / sudo

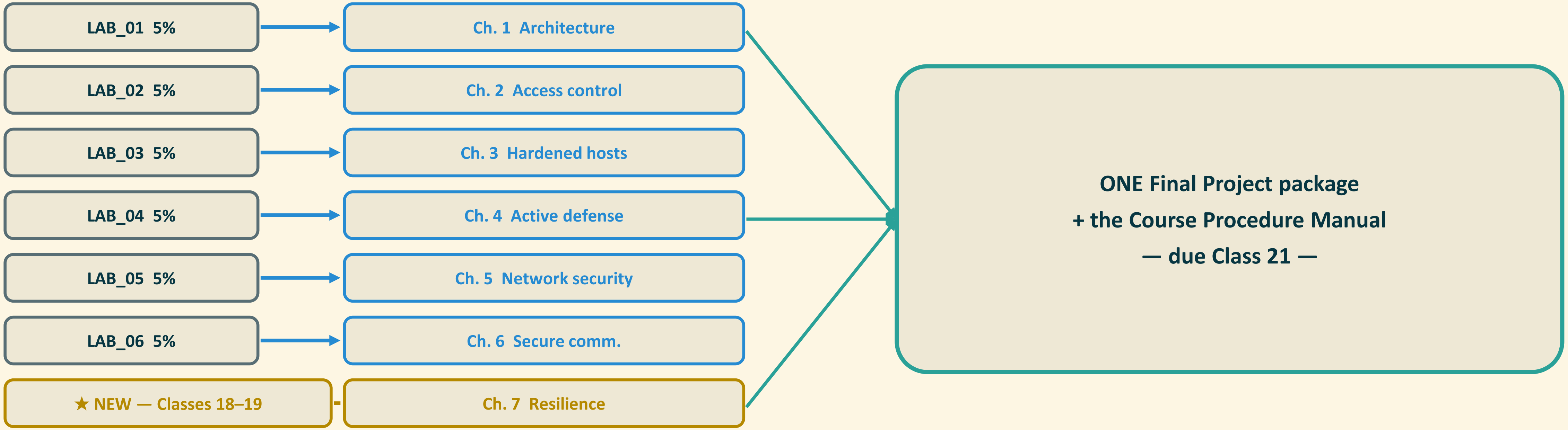
The Minimum System List — Placed in Zones

- Seven systems minimum — all virtual machines or free-tier cloud instances
- pfSense or OPNsense guards the single perimeter — everything crosses it
- The public website lives in the DMZ (Demilitarized Zone) — never inside
- Guest Wi-Fi gets its own zone (added at Class 13) — visitors touch nothing
- The IDS sensor watches; the remote worker enters only through the VPN



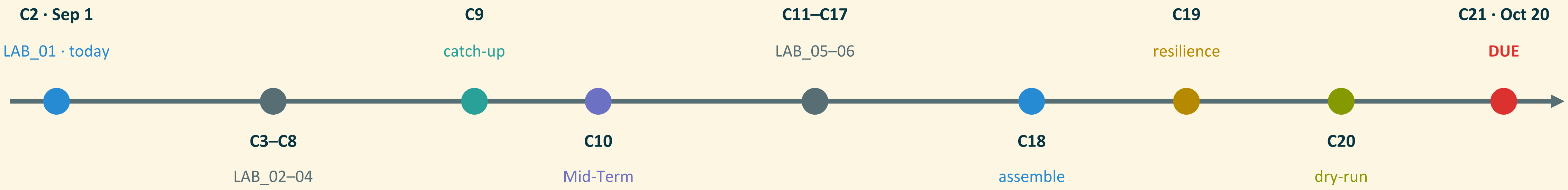
Assembly Map — Six Labs Become Seven Chapters

- Each lab (5%) becomes one project chapter — documented the SAME week
- Six labs → Chapters 1–6 · Chapter 7 (resilience) is the only new element
- The same write-ups become your Course Procedure Manual chapters
- Result: at Class 18 you ASSEMBLE — you do not start writing



Milestones at a Glance — the Road to Class 21

- Labs finish by Class 17 (October 10) — catch-up built in at Class 9
- Class 18: assembly starts · the grading checklist is re-issued UNCHANGED
- Class 19: resilience assessment · Class 20: voluntary dry-run
- Class 21 — Tuesday, October 20: BOTH deliverables due at the start of class



Chapter 1 — Defensible Architecture (LAB_01)

Must exist

- The isolated lab — set up, documented, snapshots + isolated network
- The complete network diagram: every system, entry point, and zone
- Zones: Internet · DMZ · user zone · server zone · guest Wi-Fi (Class 13)
- The zoning table — each system in exactly one zone
- Every allowed flow: source, destination, service + a 1–2-sentence reason

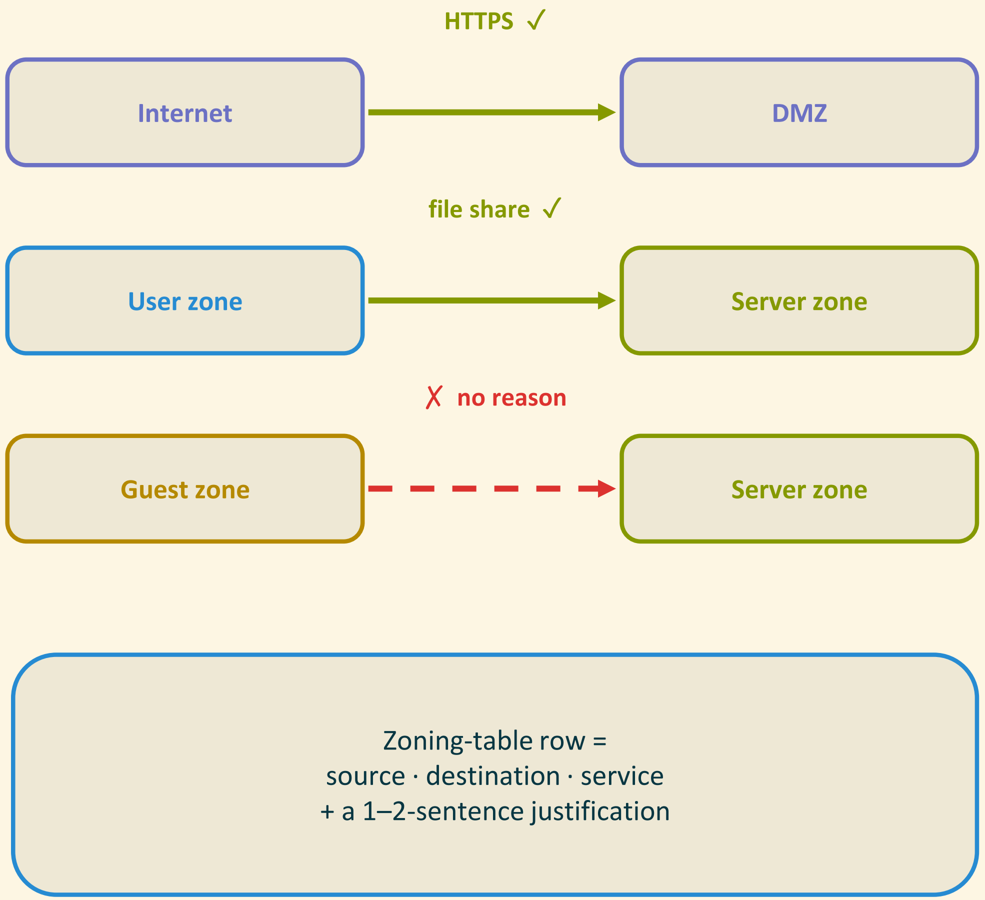
Evidence

- Finished diagram · zoning table · environment pages of the manual

Validation

- No system without a zone · no flow without a reason · trust levels stated

One arrow = one justified row of the zoning table



Chapter 2 — Access Control (LAB_02)

Must exist

- Users and groups on one Windows and one Linux machine — per the role table
- Least-privilege, need-to-know memberships: only what the role requires
- Windows: NTFS (New Technology File System) permissions and sharing
- Linux: ownership, permission bits, and a controlled sudo policy

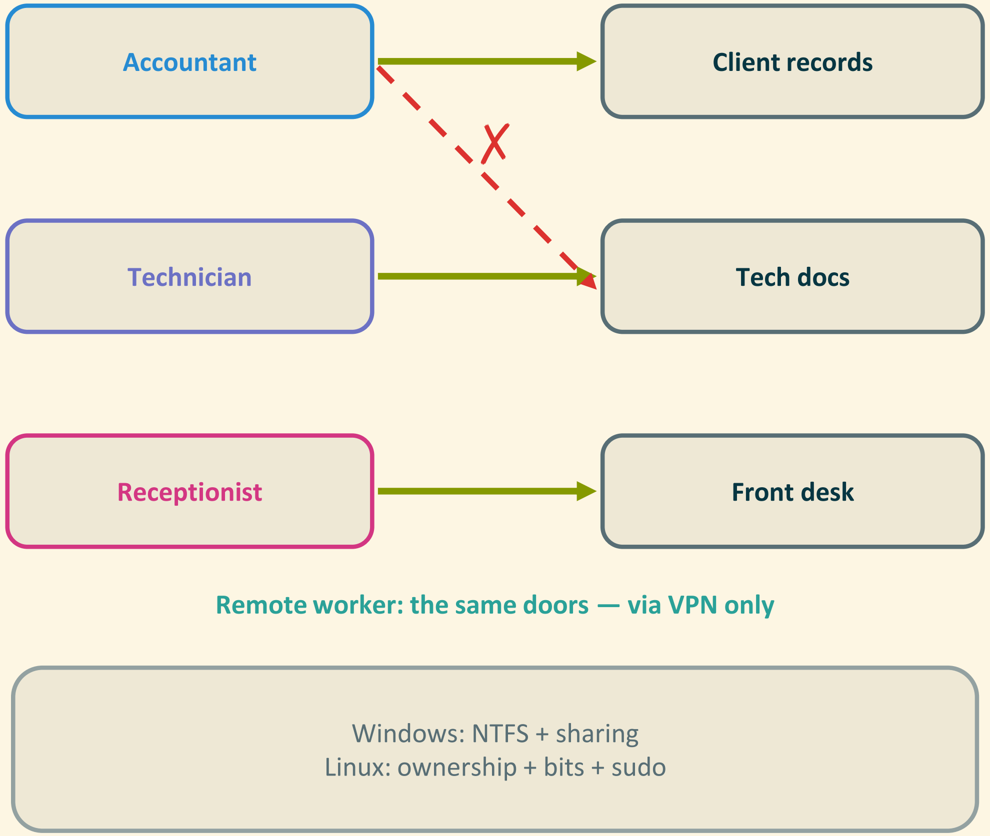
Evidence

- Permission tables · validation checklist · test-login screenshots

Validation

- Each role opens its own data — and is refused everywhere else

Least privilege — only your own door opens



Chapter 3 — Hardened Hosts & Host Firewalls (LAB_03)

Must exist

- ~10 hardening items per system: accounts, services, updates, policies
- Windows and Linux both — checked against free CIS Benchmarks baselines
- A BEFORE-and-AFTER record proving every step was applied
- Host firewall per role — Defender Firewall / UFW or firewalld (inbound rules)

Evidence

- Checklist results with before/after · rule tables · connectivity-test proof

Validation

- Every rule proven allowed/blocked by a simple test — results recorded

BEFORE

extra services running
default settings kept
updates pending · no host firewall

≈ 10 checklist steps
+ CIS Benchmarks check

AFTER

only needed services
policies applied · updated
inbound firewall rules ON

Chapter 4 — Active Defense: Intrusion Detection (LAB_04)

Must exist

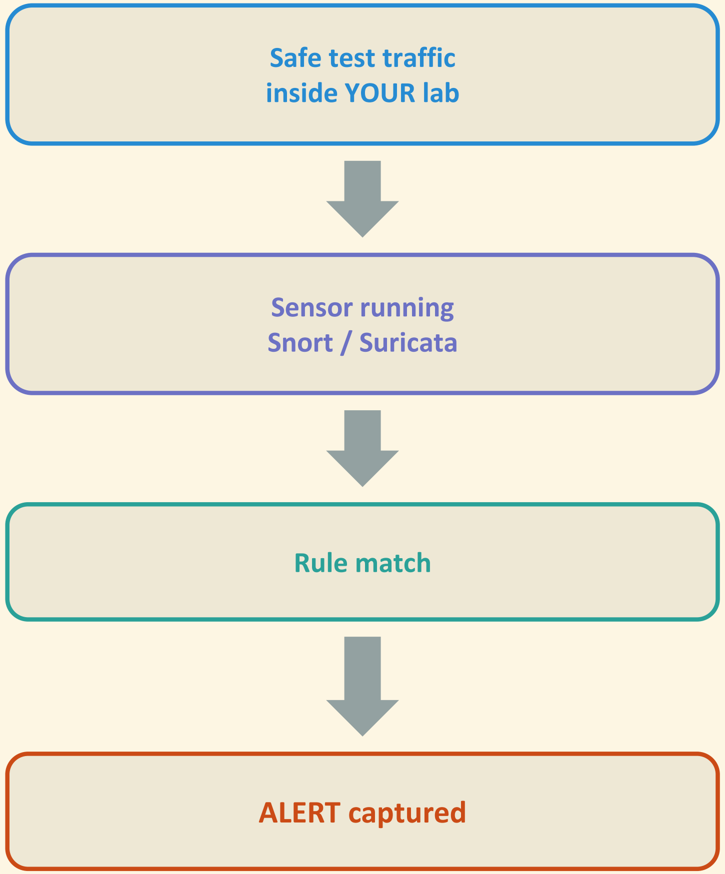
- One free, open-source IDS — Snort or Suricata, your choice
- Installed, running, and confirmed to actually SEE traffic
- The provided rule set enabled — your own simple rules are welcome
- Harmless test traffic — generated inside YOUR lab only

Evidence

- Installation notes · the rule set used · alert screenshots

Validation

- At least ONE captured alert from your own safe, contained test



Chapter 5 — Network Security: Firewall & Registry (LAB_05)

Must exist

- pfSense or OPNsense as a VM in front of the company network
- Management secured: strong credentials · HTTPS · SSH only where needed
- Default credentials removed · firmware/version recorded
- Rules that ENFORCE the Chapter 1 zoning table — flow by flow
- The device registry: role, model/VM, version, access, update status

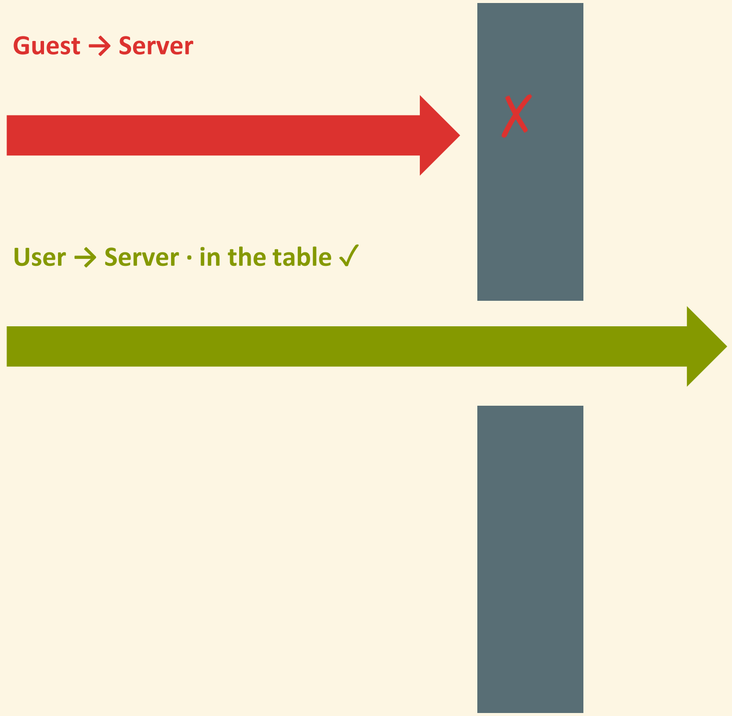
Evidence

- The rule set · the device registry · allowed/blocked proof per rule

Validation

- Every rule tested against the zoning table — allowed or blocked

The firewall enforces the zoning table



Every rule → one allowed / blocked test

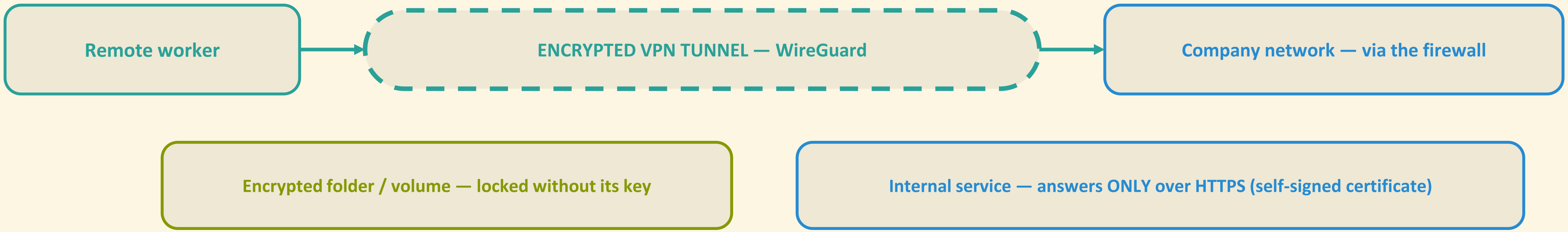
Chapter 6 — Secure Communication: VPN & Encryption (LAB_06)

Must exist

- Remote-access VPN — WireGuard recommended (OpenVPN / built-ins accepted)
- The remote-worker client connected through the tunnel — and verified
- One folder or volume of company data encrypted — data at rest
- HTTPS + self-signed certificate on one internal service — PKI in practice

Evidence & validation

- VPN config · encryption steps · certificate setup · proof for each
- Tunnel encrypted · service answers ONLY over HTTPS · volume locked without its key



Chapter 7 — Resilience Assessment: the Only New Element

- A short, template-based checklist — completed in Classes 18–19
- Ten common attack scenarios against YOUR defended company
- For each: which defense layers (Chapters 1–6) respond
- Verdict: Withstands · Partly · Does not withstand
- Plus ONE plain-language sentence of justification per row
- This is how the course assesses and documents network resilience
- Educational exercise on a fictional company — not professional consulting

10

scenarios · one sentence each

The Resilience Template (1/2) — Scenarios 1–5

- Fill EVERY row — layers that respond + verdict + one-sentence justification
- All scenarios happen inside the fictional company only

1	A phishing email plants malware on one user workstation	Withstands	Partly	Does not
2	Repeated password guessing against the remote access	Withstands	Partly	Does not
3	The public web server in the DMZ is compromised	Withstands	Partly	Does not
4	A worm tries to spread across the internal network	Withstands	Partly	Does not
5	A company laptop with client data is lost or stolen	Withstands	Partly	Does not

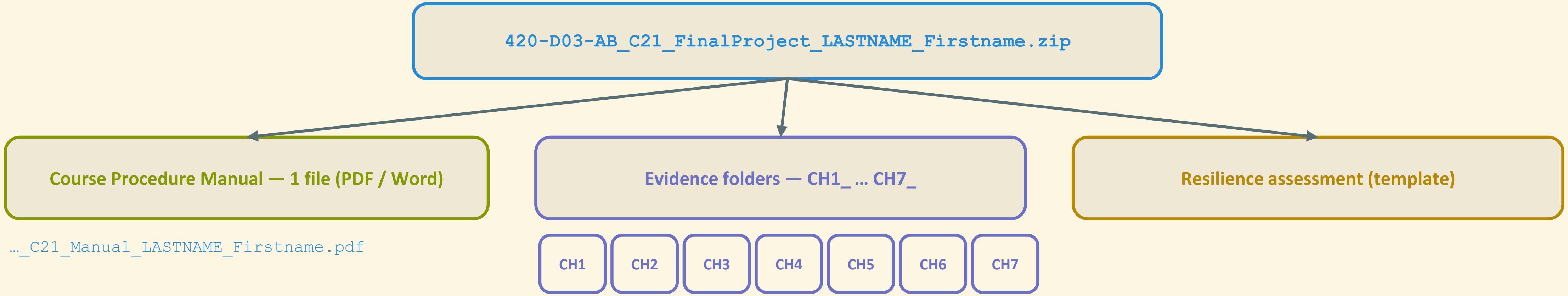
The Resilience Template (2/2) — Scenarios 6–10

- The second half of the template — same three columns to complete
- Honesty beats optimism: a justified "Partly" shows real understanding

6	A remote worker's traffic is intercepted on public Wi-Fi	Withstands	Partly	Does not
7	A DoS (Denial-of-Service) attempt targets the public service	Withstands	Partly	Does not
8	An accountant tries to open the technicians' and admin data	Withstands	Partly	Does not
9	An attacker tries default credentials on the network devices	Withstands	Partly	Does not
10	Ransomware attempts to encrypt the file server	Withstands	Partly	Does not

What You Submit — One Package, Named Exactly

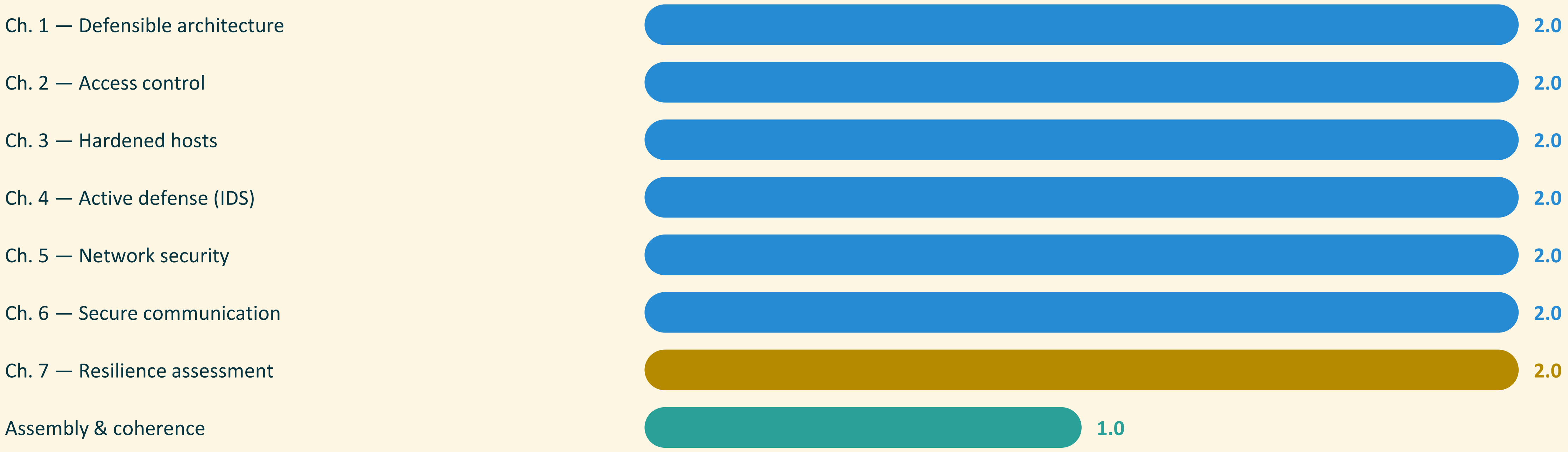
- ONE archive on the course platform, at the start of Class 21
- Inside: the manual — ONE file, PDF (Portable Document Format) or Word
- Evidence folders CH1_architecture/ … CH7_resilience/ (screenshots + exports)
- The completed resilience assessment (own file, or the manual chapter)
- AI disclosures live inside each manual chapter — nothing separate
- The lab itself stays on your machine — keep it intact until grades are out



Grading Checklist — Final Project (15 points)

- Public today — re-issued UNCHANGED at Class 18: no hidden expectations
- Each chapter: 1.0 it exists & works · 0.5 it is validated · 0.5 it is justified
- Chapters 1–7 at 2.0 points each · assembly & coherence: 1.0

! **A missing AI disclosure where AI was used = an academic-integrity issue**



Total = 15.0 points = 15% of the course grade

AI & Creativity Rules — Use It, Disclose It, Verify It

✓ You MAY use AI (Artificial Intelligence) tools at ANY stage

- Condition: each chapter states the LLM used + the EXACT prompt(s), verbatim
- This is the course's plagiarism-avoidance rule — it protects you

✓ You may substitute equivalent open-source tools — state the substitution

✓ Your own small helper scripts are welcome — and expected

- Verify before you apply: test every AI output inside your isolated lab first
- You remain fully and solely responsible for your own actions

✗ AI used without its disclosure → handled under the college's policies

LLM +

exact prompts — in every chapter

SECTION 2

The Course Procedure Manual

Due Class 21 — Tuesday, October 20, 2026 · inside the project package

10%

of your course grade

Purpose — the Manual a Real Technician Would Leave

- The professional documentation of your six labs
- Written for a colleague who was NOT there
- Every procedure must be reproducible — exactly, step by step
- The self-test: a classmate could rebuild your defended company …
- … inside their own isolated lab, from your manual ALONE
- If yes — the manual has done its job (and earns its 10%)

TEST

could a classmate rebuild it?

The Golden Rule — Write It DURING the Course

- One chapter per lab — written the SAME week the lab is done
- Use the supervised work time built into every class:
  - Weekday classes — after the 7:30 pm break
  - Saturday classes — after the 10:30 am break
- Ponderation 3-3-3: lecture · practical work · homework — 3 h each, weekly
- Class 9 is the built-in catch-up point for Chapters 1–4
- Never leave the writing for the end — that is the whole trick
- Tonight, Chapter 1 of YOUR manual officially begins

3-3-3

write during class time

Mandatory Front Matter — the Cover Pages

In this order

- 1. Cover page — course, program, competency, title, your name, session, date
- 2. The mandatory notice box — copy its text EXACTLY
- ("educational laboratory work … isolated environment … Legal Notice v1.0")
- 3. The copyright line — your procedures © you · templates © Cyber.SoHo
- 4. Short description of the fictional company (+ any renaming you made)
- 5. Table of contents

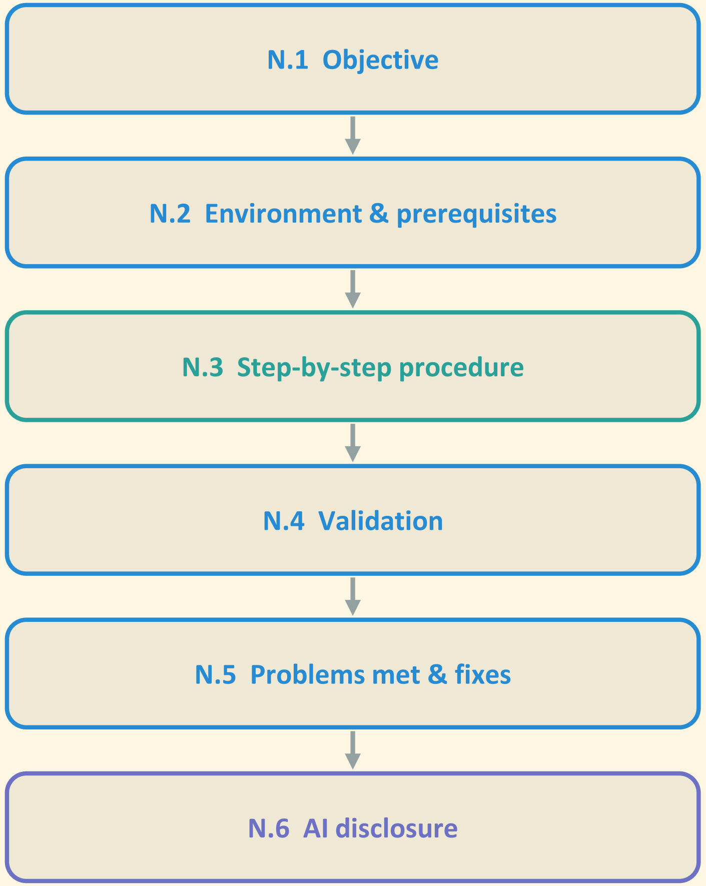
- The exact texts, ready to copy, are in the FULL document — § 2.3

EXACT

copy the notice box word for word

The Standard Chapter Template — N.1 to N.6

- Every lab chapter uses the SAME six-part skeleton
- The template file is re-distributed at Class 18 — identical
- "Nothing went wrong" is an acceptable N.5 entry



The AI Disclosure Table — One Row per Use

- Required in EVERY chapter where an AI tool helped
- The prompt goes in VERBATIM — exactly as typed
- No AI in a chapter? Write exactly: "No AI tools were used in this chapter."

#	AI tool (LLM) + version	Exact prompt given (verbatim)	Where the output was used	How verified before use
1				
2				

No AI in a chapter? Write, word for word: “No AI tools were used in this chapter.”

The Documentation Standard — 8 Rules, Every Page

- 1. Plain, professional wording — short sentences, no filler
- 2. Numbered steps for every procedure
- 3. One screenshot per key step — cropped, one-line caption
- 4. Every abbreviation expanded in round brackets on first use

✗ 5. NO real personal data and NO real credentials — ever (redact!)

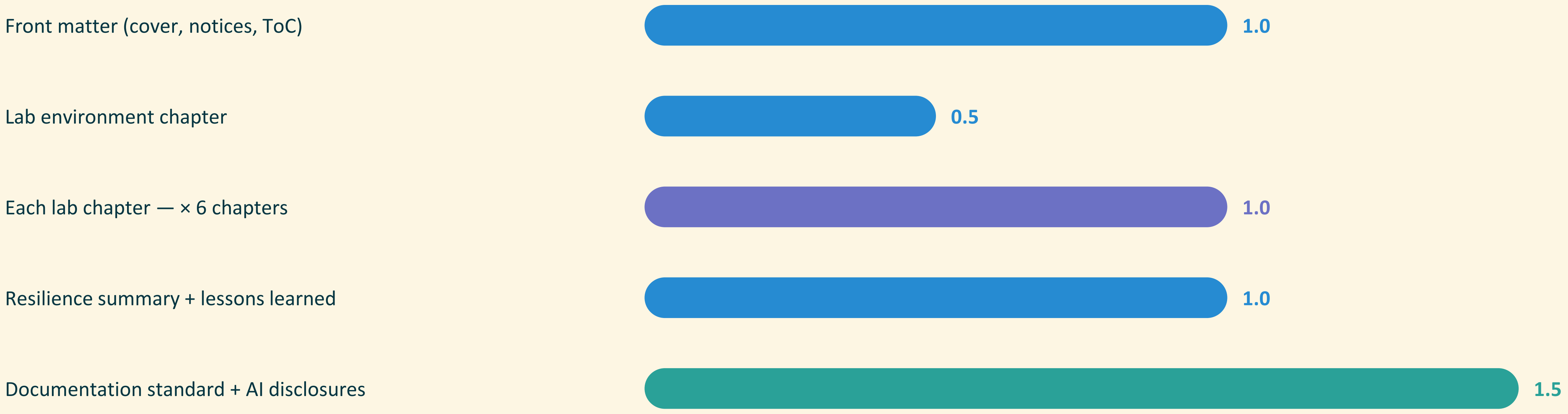
- 6. Consistent naming across chapters — matching the Chapter 1 diagram
- 7. Each chapter states the date its procedure was performed
- 8. External links: convenience only — verify first (VirusTotal URL scanner)

8

rules — on every page

Grading Checklist — Course Procedure Manual (10 points)

- Submitted as ONE file, inside the Class 21 project archive
- Complete-per-template AND reproducible — that is what each chapter point buys
- The documentation standard + every AI disclosure carry 1.5 points alone



Six lab chapters at 1.0 each = 6.0 · Total = 10.0 points = 10%

Why the Workload Stays Manageable

- The manual is written DURING the course — never at the end
- The project adds only assembly + one short checklist
- All technical work is already done by Class 17
- Four full sessions (18–21) are reserved for completion
- Voluntary dry-run against the checklist at Class 20
- 3-3-3 ponderation: keep the weekly rhythm → finish inside scheduled time

25%

secured by Oct 20 — if you keep up

Questions ?

Tonight, before you leave:

Sign today's session	Quiz Q_01 · finish LAB_01	Open Chapter 1 of your manual

The one unbreakable rule: isolated lab only — when in doubt, STOP and ask

Single source of truth: the FULL document — Final Project and Course Procedure Manual

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